Solar-Powered UAVs: A systematic Literature Review

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Abstract*— Solar-powered Unmanned Aerial Vehicles (SPUAVs), commonly known as solar drones, are an innovative and eco-friendly category of aircraft that rely on solar energy as their primary power source. Outfitted with solar panels, these drones capture and convert sunlight into electricity, substantially extending their flight durations. SPUAVs have garnered significant attention across a diverse array of applications, spanning surveillance, environmental monitoring, agriculture, disaster management, communication, weather tracking, and fire detection. By harnessing solar power, they offer compelling advantages, including greatly prolonged flight endurance, reduced reliance on fossil fuels, and cost-effectiveness. Capable of reaching altitudes exceeding 70,000 feet, they are well-suited for both civilian and military purposes. Transitioning away from non-renewable energy sources, SPUAVs present a sustainable and economically attractive alternative, uniting traditional aircraft, and unmanned aerial vehicles. Advances in renewable technologies, particularly in solar cells, rechargeable batteries, and electric motors, are revolutionizing the UAV landscape, enabling SPUAVs to achieve hitherto unattainable flight capabilities. The future portends transformative changes in aerial operations, marked by reduced environmental impact and heightened efficiency through solar-powered innovation.

Keywords*— *Solar powered UAV (SPUAV), High Altitude Long Endurance (HALE), Solar Energy, Sustainable Flight, MPPT, Solar Powered Flight, Airframe Design

I. Introduction

Energy sources for vehicles like aircraft, buses, and cars have historically relied on non-renewable fossil fuels. Aircraft serve diverse civilian and military roles, while Solar-Powered Unmanned Aerial Vehicles (SPUAVs) find applications in communications, fire detection, surveillance, and weather monitoring. Recent technological advancements have brought the once-fantastic notion of continuously flying SPUAVs closer to reality, powered by innovations like high-energy-density batteries and flexible solar cells. The concept is straightforward: solar cells cover the UAV's wings, harnessing energy during the day to supply electronics and propulsion, and recharge onboard batteries. New technologies such as solar cells, rechargeable batteries, and electric motors have transformed UAV capabilities, resulting in extended flight endurance and high-altitude operation. SPUAVs now range from minutes to months, even up to a year, reaching altitudes around 70,000 feet. These aircraft offer cost-effective, flexible geostationary satellite-like functionality, with self-launch and easy maintenance features [1], [2], [3].

II. Background and Literature Review

A. The SPUAVs System Working Principle

The basic concept is to operate the UAV using solar energy, as shown in Fig. 1. This can be accomplished by covering the parts of the UAV that are exposed to sunlight with solar cells connected in a certain configuration [6].

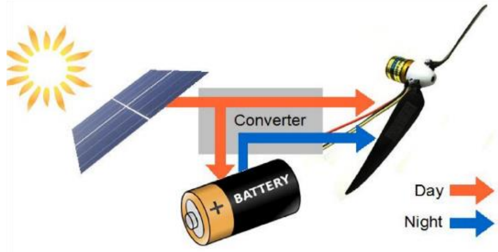


Fig. 1. Working Principle of Solar-Powered System [6].

During the day, these cells convert radiant energy (photons) into electrical energy depending on the inclination of the rays and the sun's radiation. This electrical energy is used to power the propulsion system and electronics and store the excess in the battery. The maximum efficiency of the solar cells is obtained through the use of MPPT [7]. During the night, because there is no power coming from the solar cells, the battery is discharged in order to continue operating the energy-consuming components [6].

B. Studies on SPUAVs

Researchers at San José State University explored solar-powered UAVs for extended flight. Montgomery and Mourtos designed the Photon UAV with a 1.5 m² area and 5 kg gross weight, featuring Panasonic NCR18650B lithium-ion batteries and SunPower A-300 photovoltaic panels [4]. Bhatt's team investigated achieving over 24-hour solar-powered flight, optimizing endurance with high-energy-density Lithium-Sulfur batteries, projecting a surplus 75 kW power generation [8]. Najafi's study aimed at reducing radio wave propagation through a HALE UAV as a solar communication beacon, mirroring the Helios SPUAV's 29,260 m altitude and a target of over 30 days in flight [9]. Luca Curcio focused on low-altitude solar-powered UAVs for bushfire monitoring and data transmission [10]. These studies contribute to the development of sustainable and long-endurance solar UAVs.

C. Different SPUAVs Projects

A number of projects have been conducted on SPUAVs starting in 1974 and continuing to the present day. This section will provide an overview of some of those projects.

In California in 1974, brothers Roland Boucher and Robert Boucher made the first solar-powered flight with their Sunrise

I, as shown in Fig. 2. The purpose of its construction was to study the feasibility of perpetual flight and flying using solar energy to replace expensive aircraft and satellites. Over 1000 solar cells are installed on the wings of the aircraft to produce 450 watts of power [11]. In those days, storing energy for night flights was very difficult due to the weight of the batteries. Instead, it was set up to climb during the day and descend during the night, but it failed [4].

A year later in 1975, the brothers made another flight experiment with an improved version of Sunrise I called Sunrise II as shown in Fig. 3. Sunrise II contained approximately 4,480 solar cells producing 600 watts and weighed only 1.8 kg [12].

Fig. 2. Sunrise I [9].

Fig. 3. Sunrise II [12].

The US government provided funding for NASA to launch the Environmental Research Aircraft Sensor Technology (ERAST) programme. The Pathfinder was the first high-altitude solar aircraft, and the ERAST programme used it. In 1993, it made its first flight from Dryden. It had a wingspan of 30 meters and weighed 254 kg. On September 11, 1995, it ascended to a height of 15.4 km at NASA's Dryden Flight Research Center (DFRC), and two years later, in 1997, it ascended to a height of 21.5 km at the US Navy's Pacific Missile Range Facility (PMRF) on the Hawaiian island of Kauai [14]. The development of the ERAST programme continued until 2003.

In August 1998, a new version of Pathfinder called Pathfinder Plus was launched. It reached an altitude of 24.4 km, had a wingspan of 36.9 m, and new propulsion, aerodynamic, solar, and system technologies [15]. Centurion could then carry 45 kg of data collection equipment and remote sensing for the use of scientific studies of the Earth's environment thanks to its doubled wingspan, which increased to 63.1 m. It could fly for 2 to 5 hours after sunset thanks to a lithium battery, but it wasn't able to do so all night [16]. The final prototype of the series was Helios. In 2001, Helios completed its goal which is reaching an altitude of 29.524 km and a 40 min flight near Hawaii above 29.261 km. But when Helios fell into the Pacific Ocean on June 26, 2003, some structural flaws caused it to be destroyed [13].

In 2008, André Noth, a doctoral candidate at ETH Zurich, presented a new model for solar-powered aircraft with perpetual endurance. This model, known as Sky-Sailor, Fig. 4, was a versatile design that could be applied to various types of solar aircraft. Noth's validation of the model involved the creation and flight of an airplane weighing 2.6 kg and boasting a wingspan of 3.2 m [13]. The solar panels utilized were RWE-S-32 panels, which had a power efficiency of 16%, slightly lower than the SunPower A-300 panels, but offered a superior power-to-weight ratio. Rather than attaching the solar panels to the wing's outer surface, they were integrated into the wing structure itself. Similar to Solong, Sky-Sailor employed lithium-ion laptop batteries [4].

Fig. 4. Sky-Sailor [4].

Solar endurance flight had faced challenges, with a 2007 attempt failing due to battery drain from turbulent night winds [13]. However, Sky-Sailor succeeded on June 21, 2008, completing a 27-hour flight with a rechargeable battery and a custom autopilot system maintaining altitude between 200 m and 400 m [17]. The Zephyr S, a high-altitude pseudo-satellite (HAPS), reached new heights. Its Amprius lithium-ion batteries with 100% silicon anodes offered high specific energy. Flexible, efficient solar cells from MicroLink Devices powered the aircraft. Achieving records like 25 days of flight and 21.34 km altitude, the Zephyr S excelled in communication and surveillance. UV radiation susceptibility impacted carbon fiber airframes but didn't overshadow the aircraft's success, highlighting HAPS potential for enduring services with a 25-meter wingspan and 750-kilogram weight carrying up to 15 kilograms of payload [18].

Fig. 5. Zephyr S [19].

The PHASA-35, developed by BAE Systems and Prismatic, is a solar-powered UAV designed for stratospheric operations, providing a cost-effective alternative to satellites with the maneuverability of an aircraft Fig. 6. With a 35 m wingspan, it employs a combination of solar cells and batteries for power [20].

Fig. 6. PHASA-35 [21].

The solar cells generate electricity during the day, while the batteries ensure nighttime operation. Although it theoretically boasts a year-long endurance, practical expectations range from several weeks. Its applications span various sectors, including surveillance, communications relay, disaster relief, border patrol, and environmental monitoring [21]. Technologically, it features a lightweight solar cell array, durable

batteries, and an autonomous flight control system. The PHASA-35 has achieved significant milestones, such as a successful maiden flight, over 24 hours of endurance, and an altitude exceeding 19.81 km. The UAV employs gallium arsenide solar cells for high efficiency, lithium-ion batteries for weight reduction, and relies on a GPS and INS for flight control [20].

TABLE I. REVIEW ARTICLES AND REFERENCE WORKS ON HAPS SYSTEMS [23]

NO	Author	Focus	Description
1	[24]	Wireless architecture	Review the practical specifications and possible architectural designs for the deployment of HAPS. The study is based on examples of HAPS system usage from the past.
2	[25]	Wireless architecture (Book)	Outlines the key theories supporting the use of HAPS in place of telecommunications services.
3	[26]	Wireless architecture	An overview of potential connections between HAPS and terrestrial and satellite networks.
4	[27]	Wireless architecture and communica tions links (Book)	The foundations of HAPS systems are covered, as well as the technical prerequisites for using HAPS for broadband communication. Additionally, it displays a schedule for HAPS constellations in future networks.
5	[28] and [29]	Wireless architecture and communica tions links	These studies go over the difficulties and potential uses of airborne platforms in mmWave communications. These studies primarily focus on low-flying vehicles (UAVs).
6	[30]	Wireless communica tions	The study focuses on using HAPS for wireless communication in rural Africa without access to broadband as a replacement for terrestrial systems.
7	[31]	Wireless architecture	It considers HAPS's compatibility with both current and future networks, including 5G and those beyond that. Underserved and sparsely populated areas are regarded as In their paper, they take into account the energy management of SUAV HAPs for the provision of wireless communications services in equatorial regions and areas higher up in the northern hemisphere. It was found that by doubling an airplane's wings, it could carry six times as much cargo.

To ascertain whether the UAV configurations could complete an ongoing mission, simulations of the devices were performed in various parts of the globe. The study was broken up into two distinct stages. Phase 1 involved simulating and custom-designing the average UAV's (AVUAV) 12-month operation for each of the 12 cities. Phase 2 involved simulating a 12-month flight that could have taken place in any city on the planet. A HALE UAV created at the Institute of Technology Bandung, [32] used a hybrid power source called G-HALE. The HALE UAV was created to carry out surveillance and communication missions. The UAV had to be able to take off and land at a maximum weight of 50 kg, travel at a maximum altitude of 40,000 feet and last for 12 hrs. [33] talked about how to figure out how many solar panels and batteries are needed for the HALE UAV Institut Teknologi Bandung (ITB) to fly for more than 24 hours at 20,000 feet in his paper. To meet the needs of the UAV, various solar panel types, designs, and configurations were examined. With a generated power per weight of 436 Watt/kg and a generated power per area of 224 Watt/m^2, crystalline silicon solar cells outperformed CIGS thin film, polycrystalline silicon, and amorphous silicon thin film in terms of efficiency. The total cell weight of the Monocrystalline Silicone type solar cell was 5.31 kg and the minimum area required was 5.94 m^2.

III. METHODOLOGY

A. System Requirements

1) Solar Cells

All SPUAVs are equipped with ultra-lightweight photovoltaic cells, so they can fly at nighttime due to the stored power during the daytime [38]. During the day, Zephyr-s flies at high altitudes to maximize sun irradiation, and at night, descends to a low altitude to conserve energy [32]. A UAV powered by renewable energy must have a power management system, thin-film solar arrays, fuel cells, and electrolysers to ensure long-term endurance. Developing efficient systems for gathering, controlling, storing, and using energy is the main challenge that solar must overcome [5]. The following table provides a brief overview of the types of photovoltaic cells commonly used in SPUAV projects to assist in selecting the most suitable type of cell for the project.

TABLE II. THE MAIN TYPES OF PHOTOVOLTAIC CELLS USED IN SPUAV PROJECTS [33] AND [39]

Comparison	Monocrystalline Silicon	Polycrystalline Silicon Cell	Thin Film Cells
Examples	SunPower C60 Xinpuguang PV	Kyocera 305W JA Solar 300W	Amorphous silicon CdTe (cadmium telluride)
Efficiency (%)	15-25%	15-17%	7-13%
Shape	Pseudo-Square	Square	Square Rectangular circular
Advantages	• High power output • 25+ year lifetime	• Low cost • Less waste	• Cheaper

		• 25-year lifespan	• available in a variety of sizes • flexible
Disadvantages	• Expensive • Slightly less affected by high temperatures compared to polycrystalline panels.	• Lower space efficiency • Shorter lifespan	• Shorter lifespan • Low space efficiency • Lowest efficiency

Compared to other types of photovoltaic cells, monocrystalline silicon photovoltaic cells have the highest efficiency ratio between 15% and 25%. The monocrystalline silicon photovoltaic cells are also the most suitable for project work conditions since they are less affected by high temperatures as well as having the highest power production capacities [33] and [39]. Table IV compares the properties of three examples of monocrystalline silicon cells.

TABLE III. COMPARISON OF EXAMPLES OF MONOCRYSTALLINE SILICON CELLS [40]

Examples	SunPower C60	Xinpuguang PV	Sunyima
Efficiency (%)	21.8 - 22.5	20.6	20.2
Power output (W)	3.06	2.5	2.84
Efficiency (W/m2)	224	196	-
12-hours Efficiency (W*h/m2)	1175.36	1028.44	-
Weight (kg)	0.008	0.008	0.008
Size	125 x 125 mm	125 x 125 mm	125 x 125 mm
Price	$4.68	$1.30	$1.20
Warranty	25 years	25 years	10 years

According to the results of the comparison in the previous table, SunPower C60 cells were selected since this project requires the highest efficiency and the highest power output. For designing and analyzing the performance of SunPower C60 solar cells, Fig. 7 and Table V illustrate the cell shape and specifications in detail.

Fig. 7. SunPower C60 Solar Cells [19].

TABLE IV. SUNPOWER C60 SOLAR CELLS SPECIFICATIONS [41] AND [42]

Parameter	Value	Unit
Length and width	0.125 × 0.125	m
Mass of solar cell	0.008	Kg
Efficiency of solar cell	21.8% - 22.5%	-
Area of single solar panel	0.015	m2
Peak current (IMPP)	5.37	A
Peak voltage (VMPP)	0.57	V
Short circuit current (I_{SC})	6.24	A
Open circuit voltage (V_{OC})	0.682	V
Thickness	1.65×10-4	m
Peak power	3.06	W
Price	4.68	$

2) Batteries

Batteries are considered to be one of the vital parts of an SUAV. Their weight, energy density, and storage capacity could affect the performance of the aircraft. Batteries are divided into primary batteries that cannot be recharged and secondary batteries that can be recharged [34]. [7] mentioned in his study the characteristics of different as displayed in Table II.

TABLE V. CHARACTERISTICS OF DIFFERENT TYPES OF SECONDARY BATTERIES [7] & [35]

Characteristic	Lead-acid	NiCd	NiMH	Li-ion	Li–Po	Li–S	Zn air
Energy density	33–40 W h/kg	40–60 W h/kg	30–80 W h/kg	160 W h/kg	130–200 W h/kg	250–350 W h/kg	230 W h/kg
Energy/volume	50–100 W h/L	50–150 W h/L	140–300 W h/L	270 W h/L	300 W h/L	600 W h/L	270 W h/L
Power density	80–300 W/kg	200–500 W/kg	250–1000 W/kg	1800 W/kg	2800 W/kg	2800 W/kg	105 W/kg
Recharge time	8–16 h	1 h	2–4 h	2–3 h	2–4 h	–	10 min
Cycle eff	82%	80%	70%	99.9%	99.8%	99.8 %	–
Lifetime	–	–	–	24–36 mth	24–36 mth	24–36 mth	–
Life cycles	300	500	500–1000	1200	41000	41000	42000
Nominal voltage	2 V	1.2 V	1.2 V	3.6 V	3.7 V	3.7 V	1.2 V
Operating temperature Commercial use	Ambient (15–25 °C) 1900/1970	20 to 60 °C 1900/1950	20 to 60 °C 1990	40 to 60 °C 1991	– 1999		Ambient – 25 °C
Cost per W h	$0.17	$1.50	$0.99	$0.47	–	–	–

The Zephyr series uses the Lithium Sulfur battery, which was developed by SION Power Company. These types of batteries helped the Zephyr series have an endurance of more than 14 days [10]. As shown in Table II, the Li-S had a better performance in terms of energy density and energy/volume than the Li-ion and Li-Po. [36] mentioned that "Energy density is used to determine the efficiency of a rechargeable battery." The EAV-3 and PHASA-35 used a lithium-ion battery system. This battery type allowed the PHASA-35 to operate at an altitude above 20 km and had an endurance of 12 months. This power system could achieve 400 daily cycles at 90% capacity [35]

It should be possible to fit the battery inside the main wing [32]. The drawback of the Li-S, as [37] said, is that they "cannot be recharged enough times before they fail to make them commercially viable." The Li-ion battery was chosen for this project as the longest endurance UAV (PHASA-35). There is a considerable advantage when using lithium metal to make rechargeable batteries due to its low density, high energy-to-weight ratio, and high electrochemical potential [37]. According to the aforementioned data, lithium-ion will be utilised in solar-powered aircraft because it:

- High cycle efficiency.

- Light in weight.
- High energy density.
- And occupied a small space.

3) Maximum Power Point Tracker (MPPT)

During the SPUAV mission, the point where the maximum power is achieved is variable. To determine the percentage of energy used to produce voltage and the amount of energy used to generate current, it is important to use a device that monitors these data. Both products of both (V*I) provide the maximum amount of power to the propulsion system or battery of the SPUAV and produce the maximum amount of power as shown in Fig. 8 [43].

To maximize energy extraction in all conditions, maximum power point trackers (MPPT) are commonly used with wind turbines and solar photovoltaic (PV) systems. While the principle is commonly applied to solar energy, it can also be applied to other sources of variable power, such as optical power transmission and thermo-photovoltaic power transmission [44]. In the MPPT, the main problem is that the amount of sunlight that falls on the solar panels, the angle of incidence of the solar light, the temperature of the solar panels, and the characteristic electrical loads all affect the efficiency of the transfer of energy from the solar cells [12].

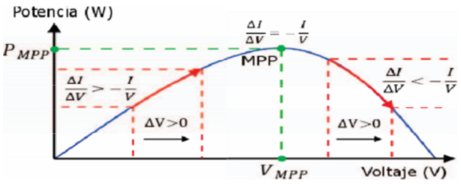


Fig. 8. Incremental conductance MPPT algorithm [45].

Fig. 9. Genasun-GV-10-Li-16.7V-Boost-MPPT specifications [46].

For this project, a commercial Genasun-GV-10-Li-16.7V-Boost-MPPT regulator Fig. 9 was selected for two reasons. According to Table VI, the selected regulator can receive a maximum of 160 watts of solar power, which is the highest rating among all Genasun regulators. The regulator chosen can handle a 4-cell system and is approved for the system, including the battery and motor [46].

TABLE VI. GENASUN-GV-10-LI-16.7V-BOOST-MPPT SPECIFICATIONS [46]

Parameter	Value	Unit
Maximum Recommended Panel Power	160	W
Rated Battery (Output) Current	10.5	A
Recommended Max Voc at STC	27	V
Minimum Battery Voltage for Operation	8.5	V
Input Voltage Range	0 - 34	V
Maximum Input Short Circuit Current	10.5	A
Maximum Input Current	19	A

B. Avionics' components

Avionics are the electronic components used to power and control solar-powered unmanned aerial vehicles (SPUAVs). They include the communication system, the flight controller and so on. Based on the size, weight, and power requirements of SPUAVs, avionics types and characteristics are determined [47].The SPUAV is propelled by motors, which convert electrical energy into mechanical energy. The brushless DC motor is the most common type of motor used in SPUAVs. Compared to brushed motors, brushless DC motors are more efficient, require fewer maintenance requirements, and do not require brushes. Motor characteristics are determined by the thrust, weight, and speed requirements of the SPUAV [48].

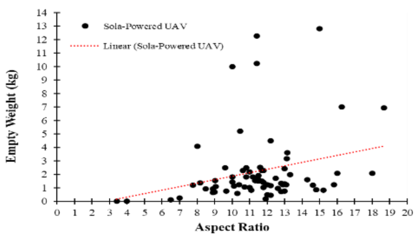
Based on the previously defined mission specifications, the system components were identified using eCalc as shown in Table VII. The eCalc provides high-quality, web-based tools for calculating, simulating, designing, and evaluating electric brushless motor drives for RC pilots of UAVs, airplanes, EDF jets, helicopters, and multi-rotors.

TABLE VII. THE SELECTED AVIONICS AND PROPULSION COMPONENTS

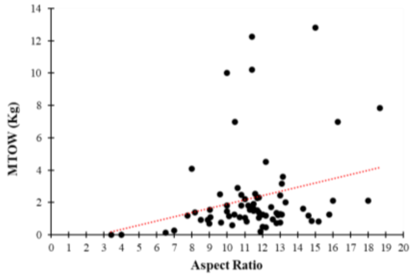
			
PROPDRIVE v2 4248 650KV Brushless Outrunner Motor	MG995 Metal Gear Servo Motor	Walksnail Avatar HD Kit V2	Readytosky 80A ESC 2-6S Brushless ESC
			
TS832 Transmitter (48CH)	Hex Pixhawk 2.1 PX4 PIX 32 Bit Flight Controller Autopilot	Folding Propeller 15x8	3DR Radio Telemetry 433mhz 1000MW

C. Airframe Design

In order to design the wing of the aircraft, the aircraft's weight must be determined first. The empty weight of current solar-powered UAVs is given in Fig. 10 (a) while the MTOW is given in Fig. 10 (b). From both figures, it can be seen that the majority of the aircraft are designed with an aspect ratio ranging between 10 to 13 with an average empty weight and MTOW of 2kg. Note that these figures are only considering small-scale UAVs, and since most of them are not designed to carry payloads, their empty weight is equal to their MTOW.



(a) Empty weight vs Aspect ratio.



(b) MTOW vs Aspect ratio.

Fig. 10. Comparison between the weights of current Solar-powered UAVs and their respective aspect ratio.

One of the important aspects of designing a solar-powered UAV is the design of its wing. The wing shall not only generate the necessary lifting force to overcome the aircraft's weight, but it should also be able to be fitted with solar cells. And since these types of UAVs usually fly at low speeds, the selected airfoils are usually in the low Reynolds number airfoil category. Examples of such airfoils are SD5060, SD7032, NACA2412, Eppler e210, and Phoenix airfoil. The currently used airfoils in some of the designed solar-powered UAVs are given in Table VIII.

TABLE VIII. CURRENT SOLAR-POWERED UAVS WITH THEIR GEOMETRIC DESIGN

UAV Name	Wingspan (m)	Wing Area (m2)	Aspect Ratio	Airfoil Type
Project Solaris (University of Sweden)	2.50	0.38	16.60	Phoenix Airfoil
Sky Saylor	3.24	0.81	13.00	WE3.55-9.3
Atlantik Solar	5.69	1.74	18.67	MH139-F
SoLong	4.75	1.5	15	–
Mini-Phantom Solar UAV	3.2	0.934	10.6	–
Solar Powered UAV (National University of Defense Technology-China)	5	1.54	16.27	SD7032
Solar Powered UAV V.1.0 (University of Minnesota)	4	1.22	13.11	SD5060
Solar Powered UAV V.1.1 (University of Minnesota)	4	1.22	13.11	Eppler e210
Sun Sailor 2 UAV	4.2	1.34	13.15	SD7032

This study highlights critical design considerations for Solar-Powered UAVs (SPUAVs), focusing on two versions, V.1.0 and V.1.1. While both share similar wingspan, area, and aspect ratio, they differ in airfoil configuration. V.1.0 employs the SD5060 airfoil due to its flat upper surface, suitable for solar cell installation. V.1.1, benefiting from flexible solar cell technology, adopts the curved Eppler e210 airfoil, enhancing aerodynamic efficiency and reducing power requirements for level flight.

Optimizing the wing's aspect ratio is essential, as depicted in Fig. 11, with aspect ratios falling within 10 to 13 and wingspans ranging from 2 to 4 meters. A balance is needed, ensuring high aerodynamic efficiency without compromising structural integrity.

Fig. 12 illustrates the relationship between wing area and aspect ratio, emphasizing the need to strike a balance. A larger wing area accommodates more solar cells but decreases aspect ratio, impacting aerodynamic efficiency. SPUAV designers must carefully consider these trade-offs to enhance performance while incorporating the latest solar technology.

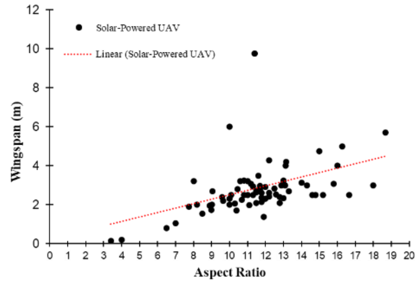


Fig. 11. Comparison between wingspan and aspect ratio for current solar-powered UAVs.

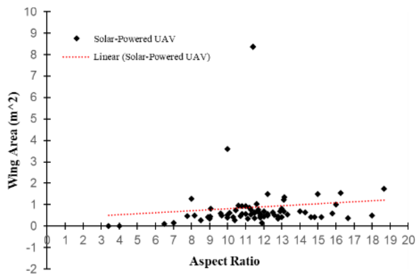


Fig. 12. Comparison between wing area and aspect ratio for current solar-powered UAVs.

D. *Airfoil Selection*

When the wing design is finished, by using the following criteria the airfoil was selected: high CL, low CD, and smaller camber for cell placement. Here, we look at two airfoils that were adapted from earlier prototypes.

TABLE IX. THE CONFIGURATION SELECTION OF THE SPUAV

Classification		Value	Unit
Main wing	Wingspan	24.2	m
	Chord length	1.10	m
	Area	30.3	m2
	Aspect ratio	18.1	–
	Airfoil	Eppler e210	–
	Taper ratio	0.26	–
Horizontal wing	Wingspan	4.20	m
	Chord length	0.80	m
	Area	3.36	m2
	Aspect ratio	5.25	–
	Airfoil	NACA 0010	–
	Taper ratio	1.00	–
Vertical wing	Wingspan	2.40	m
	Chord length	0.80	m
	Area	1.92	m2
	Aspect ratio	3.00	–
	Airfoil	NACA 0010	–
	Taper ratio	1.00	–
Fuselage length		8.15	m

The flying velocity, working density (ρ), and maximum take-off weight (WTOmax) all have an impact on the amount of L and D that the SUAV needs to fly. A well-known theoretical estimation is used to forecast (CL and CD), as shown in Eqs 4.1 and 4.2. Additionally, the wing area can be calculated using (Rajendran et al., 2017).

$$C_L = WTO_{max} / (0.5*\rho*v\ 2*S) \tag{1}$$

$$C_D = C_{DoW} + C_L^2\ (\rho*\varepsilon*S) \tag{2}$$

$$S = \mathrm{b}^2*\mathrm{AR} \tag{3}$$

IV. Discussion

The literature review on Solar-Powered Unmanned Aerial Vehicles (SPUAVs) provides an extensive examination of this innovative and eco-friendly aircraft category. These solar drones, equipped with solar panels, harness solar energy for sustainable and renewable power, significantly extending their flight durations. SPUAVs find applications in surveillance, environmental monitoring, agriculture, disaster management, and communication, offering promising solutions to increase flight endurance, reduce reliance on conventional fuels, and lower operational costs. With the ability to reach altitudes over 70,000 feet, SPUAVs bridge the gap between traditional aircraft and UAVs, suitable for both civilian and military use.

Advancements in renewable energy technologies, including solar cells, rechargeable batteries, and electric motors, have transformed the SPUAV landscape, enabling unprecedented flight endurance and capabilities. By capturing solar power during daylight and storing surplus energy for nighttime operations, SPUAVs have the potential to significantly impact aerial operations. The review traces the historical development of SPUAVs, from the Boucher brothers' pioneering solar flight to NASA's Environmental Research Aircraft Sensor Technology (ERAST) program and more recent innovations like Zephyr S and PHASA-35.

The methodology section underscores the importance of key components in SPUAV design, such as batteries, solar cells, and Maximum Power Point Tracker (MPPT) systems, in optimizing efficiency, endurance, and power generation. The review provides valuable insights into SPUAV operational principles, historical progress, and essential considerations for system design. It serves as a strong foundation for further research and development in the field, emphasizing the potential of solar-powered innovation in revolutionizing aviation and reducing environmental impacts.

V. Conclusion

The literature on Solar-Powered Unmanned Aerial Vehicles (SPUAVs) showcases a dynamic and promising field where renewable energy, aviation, and technological innovation converge. These solar drones are poised to revolutionize the aerospace industry by prioritizing sustainability. Their unique capability to capture solar energy, convert it into electricity, and store excess power in onboard batteries greatly extends their flight endurance. This quality makes them indispensable in domains such as surveillance, communication, environmental monitoring, agriculture, and disaster management. Recent technological advances, including high-energy-density batteries, flexible solar cells, and Maximum Power Point Tracking (MPPT) systems, have brought the vision of continuous solar-powered flight closer to reality. These SPUAVs can operate at high altitudes, reducing reliance on conventional fuels and minimizing environmental impact. Pioneering projects like Zephyr S and PHASA-35 exemplify the potential of these systems to provide persistent, cost-effective services. The selection of components is crucial in SPUAV design, with monocrystalline silicon cells offering high efficiency and power output, lithium-ion batteries ensuring long-endurance flights, and MPPT regulators optimizing solar power utilization. Wing design, airfoil selection, and aspect ratio are fundamental considerations, balancing aerodynamic efficiency with the installation of solar panels. The future of SPUAVs holds great promise, with innovative projects like Zephyr S and PHASA-35 driving continuous evolution and offering a sustainable, cost-effective alternative to traditional aircraft, promising transformative impacts on aviation efficiency and environmental sustainability through solar-powered innovation.

VI. Recommendations

The thorough literature review of Solar-Powered Unmanned Aerial Vehicles (SPUAVs) traces their evolution, emphasising the pivotal role of renewable energy, particularly solar power, in reshaping aviation sustainably. It covers operational principles, historical advancements, and technological innovations, serving as a foundation for further exploration in solar-powered aviation. Highlighting how SPUAVs leverage solar energy to significantly extend flight durations, it revolutionises sectors like surveillance, environmental monitoring, agriculture, and disaster management. Notably, the methodology section provides crucial insights into batteries, solar cells, and MPPT systems, crucial for optimising efficiency and endurance in SPUAV design.

The proposal suggests crafting an ultra-light, fixed-wing SPUAV with a three-meter wing span, utilising C60 solar cells for flights lasting over 12 hours. This SPUAV aims to serve both civilian and military purposes, particularly in surveillance and mapping operations.

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